# A Biologically-Inspired Framework for Online Mining Spatio-Temporal Dependency Rules from Event Data Streams

POSTER

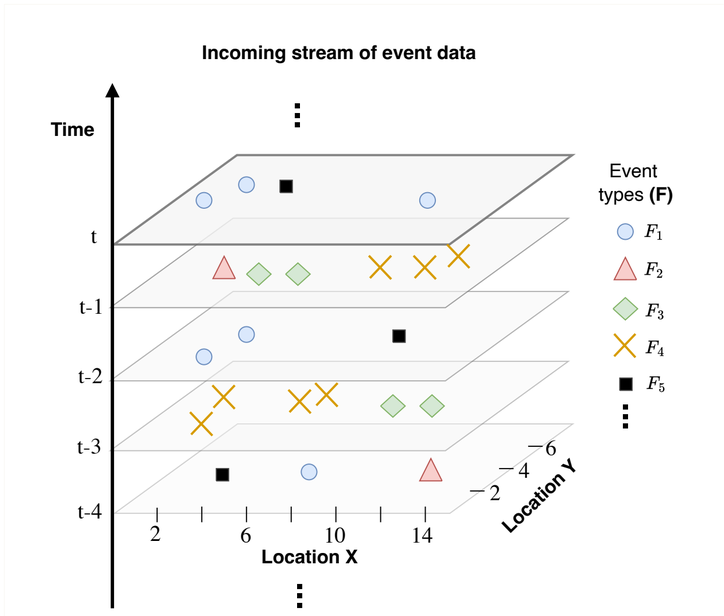
**R-205**

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## Spatio-temporal dependency rules

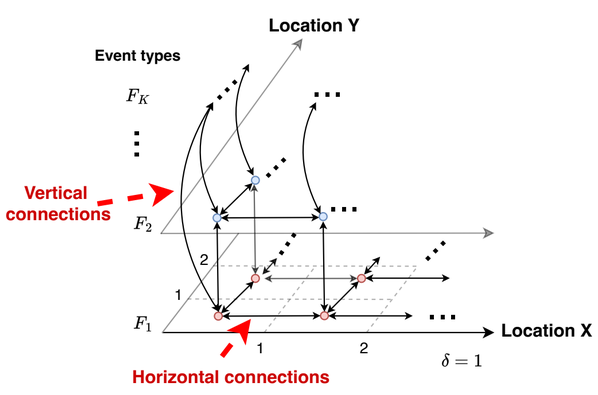



- Each event in data stream $\mathbf{D}$ has a type, location and timestamp.
- An **STDR** is a spatially annotated ordered pair of event-type/location tuples. Its weight $w_{ij}$ expresses the **significance or strength** of the dependency.
- The dependency reflects the current state of $\mathbf{D}$. These are statistical dependencies, not causal effects.

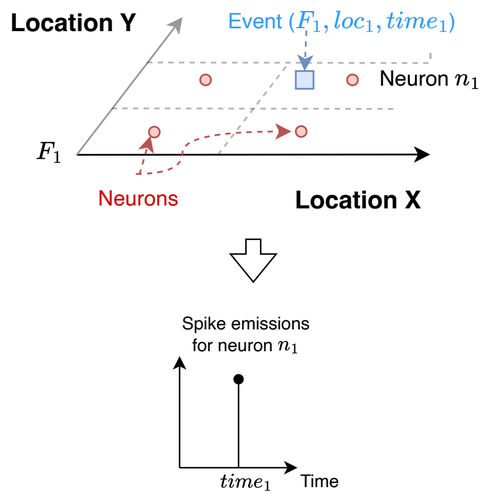
$$(F_i, \mathrm{loc}_i) \xrightarrow{w_{ij}} (F_j, \mathrm{loc}_j).$$

## Network architecture

The spatial region of stream $\mathbf{D}$ is partitioned into $m_1 \cdot m_2$ rectangular cells, with one neuron for each event type in each cell.




Neurons within each cell are fully connected by bidirectional synapses (vertical connections). Horizontal connections link neurons in cells whose centers lie within spatial threshold $\delta$.




**Event-driven neuronal model.** A neuron emits a spike at the corresponding event's timestamp.

## Learning and rule selection

Synaptic weights are initialised to zero and updated using an **unsupervised, STDP-inspired Hebbian learning rule**:

$$\frac{dw_{ij}}{dt} = \underbrace{\eta A}_{\text{reinforcement}} - \underbrace{\eta R w_{ij}}_{\text{scaling}} - \underbrace{\lambda w_{ij}}_{\text{forgetting}} .$$

The **presynaptic** $A$ and **postsynaptic** $R$ **traces** decay exponentially with timescales $\tau_A$ and $\tau_R$. $\eta$ controls learning and $\lambda > 0$ controls long-term weight decay.

For the continuous equation ($0 \le A \le 1$, $R \ge 0$), significant STDRs satisfy:

$$w_{\max} = \frac{\eta}{\lambda}, \qquad w_{ij} \ge \theta w_{\max}, \quad 0 < \theta < 1.$$

## *Online STDR miner* with lazy updates

$\mathbf{W}$ stores the currently significant synapses and their weights ($w_{ij} \ge \theta w_{\max}$).

1. For each incoming event, emit a spike by the corresponding neuron.
2. Select only the firing neuron's incoming and outgoing synapses and previously significant synapses (**lazy approach**). Decay their weights and traces using elapsed time, then apply the learning rule.
3. Prune from $\mathbf{W}$ all weights below threshold $\theta w_{\max}$. Report all significant STDRs induced by $\mathbf{W}$.

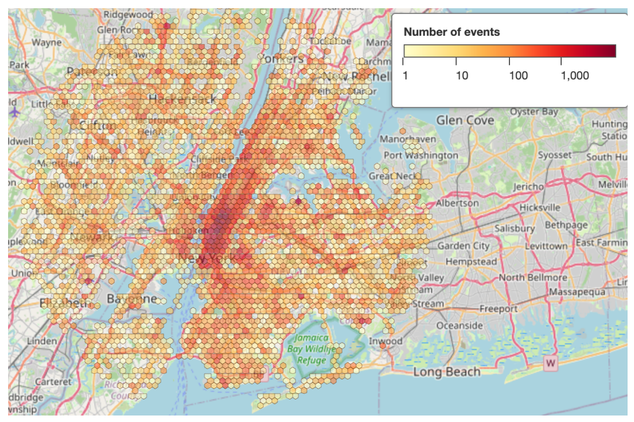
**Time per event** $O(d_{\max} + |\mathbf{W}|)$
**Memory** $O(|\mathbf{E}| + |\mathbf{W}|)$
$d_{\max}$: maximum total in- and out-degree; $\mathbf{E}$: all connections.

## Evaluation

**227,428** NYC Foursquare check-ins

**251** event types

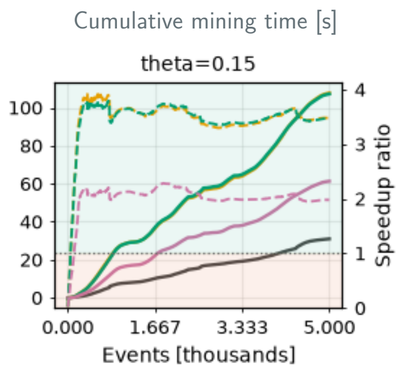
Comparison with: **STS-Miner, STBFM and Propagation Patterns** on a 120-minute moving window, with patterns restricted to pairs or two-node structures.




Density of event locations (log scale).

## Mining time and memory usage

### Online STDR miner tends to outperform the other approaches.

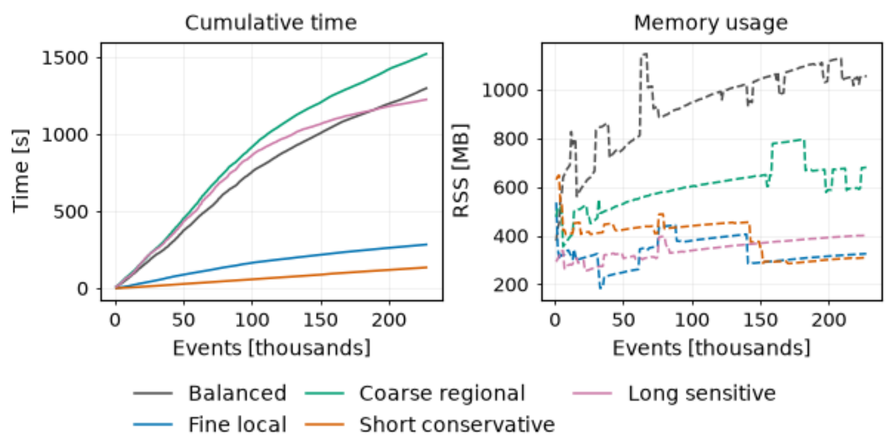



Online STDR Miner
STS-Miner
STBFM
Propagation Patterns

Solid: mining time. Dashed: baseline-to-online speedup. Ratios above one favor Online STDR Miner.

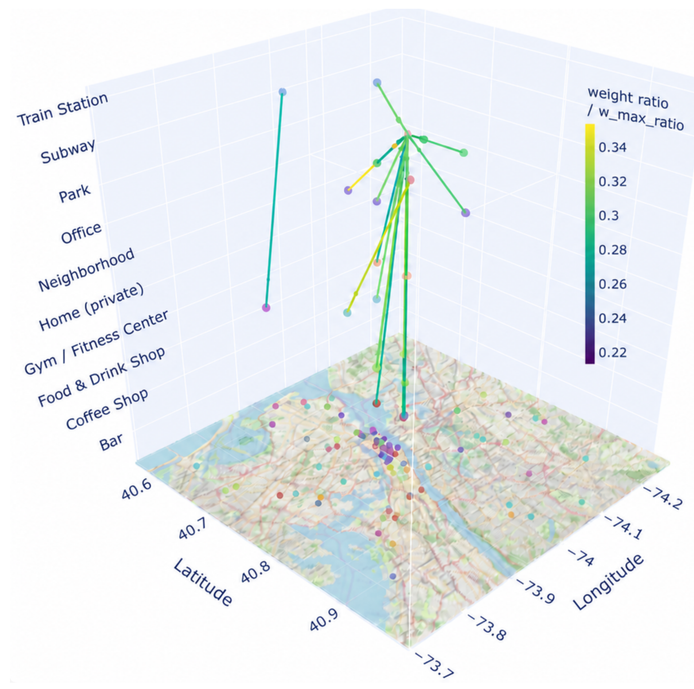
The methods have different pattern semantics.

### Runtime and memory over the full stream




All 227,428 events, with configurations differing in the spatial grid, neighborhood radius and learning parameters.

## Example STDRs




**Data stream time: 18:00–21:00, Siginficance threshold:** $\theta = 0.2$**.** The strongest reported rule links an **Office cell in Brooklyn** with a **Subway cell in Manhattan**.

## Conclusions and limitations

**Contributions.** An evolving network that represents spatially explicit dependency rules. A new learning rule with an upper bound on synaptic weights for the continuous equation.

**Current limitations.** STDRs capture only **pairwise dependencies**. Results depend on the spatial grid, neighborhood radius and learning parameters. Evaluation covers one application domain, and rule significance can be difficult to interpret.

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**Code and supplementary material**
github.com/piotr-maciag/Online_STDR_Miner